

Gas Absorption in Packed Bed

Group #2

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Introduction

The objectives of this experiment is to determine:

- The fluid flow and gas absorptions characteristics in a packed bed gas absorbed
- The overall mass transfer coefficient in the removal of CO_2 from an air stream with water in a packed bed gas absorber

Theory

- Gas absorption is a process in which one or more components of a gas stream are removed by being absorbed into a nonvolatile liquid.
- In this experiment, the absorption is physical, meaning that the driving force is that the gas removed has a greater solubility in the solvent than other gases.
- The packing in the columns provides a larger surface area for the gas and liquid to be in contact with one another so that absorption of the gas into the liquid can be increased, while also allowing even flow of liquid and gas throughout the column

Theory (cont.)

- Main equations used^[1]:

- Void Fraction

$$\varphi = \frac{V_{CO2}}{V_{Air}}$$

- **Henry's Law**

$$p_A = H_A * x_A$$

Where:

p_A = Partial pressure of component A

H_A = Henry's Constant for component A

x_A = Molar fraction of component A in Liquid Solvent

- We can use the relation:

$$y_A = \frac{p_A}{P_{tot}} = \text{Relationship of gas mixture and partial pressure of A}$$

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Theory (cont.)

- The relationship of Mass Transfer coefficient can be related to the relationship of Henry's Law as follows^[2]:

$$\frac{N_A}{a} = K(x_{A1} - x_{A2})$$

Where:

N_A = Moles of component A

a = Surface area of Water

K = Mass transfer coefficient

x_{A1} = Mole fraction of A at Point 1

x_{A2} = Mole fraction of A at Point 2

Theory (cont.)

This is rearranged as the following:

$$N_A = Ka\Delta C$$

Where delta C will include the difference in mole fractions, concentration profile from different points, and partial pressure drops. Since the experiment consists of reaching equilibrium of the gas and liquid mass transfer rates, then we can conclude:

$$Ka_{gas} = Ka_{liquid}$$

To know the levels of CO₂ in water, the concentration of CO₂ was titrated with NaOH as the base and can be determined with Equation 6:

$$C_{CO_2} = \frac{V_{base} * C_{base}}{V_{water}}$$

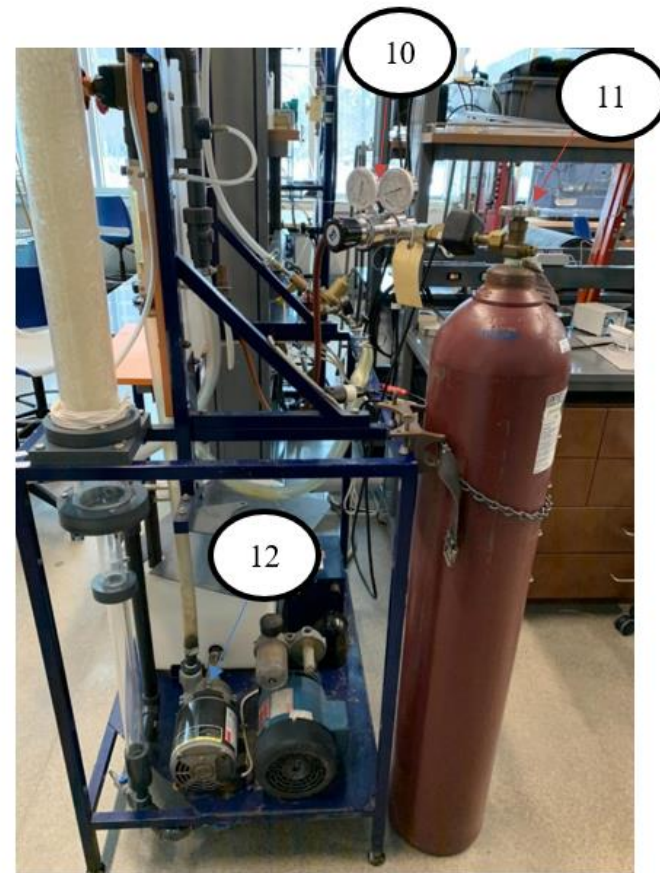
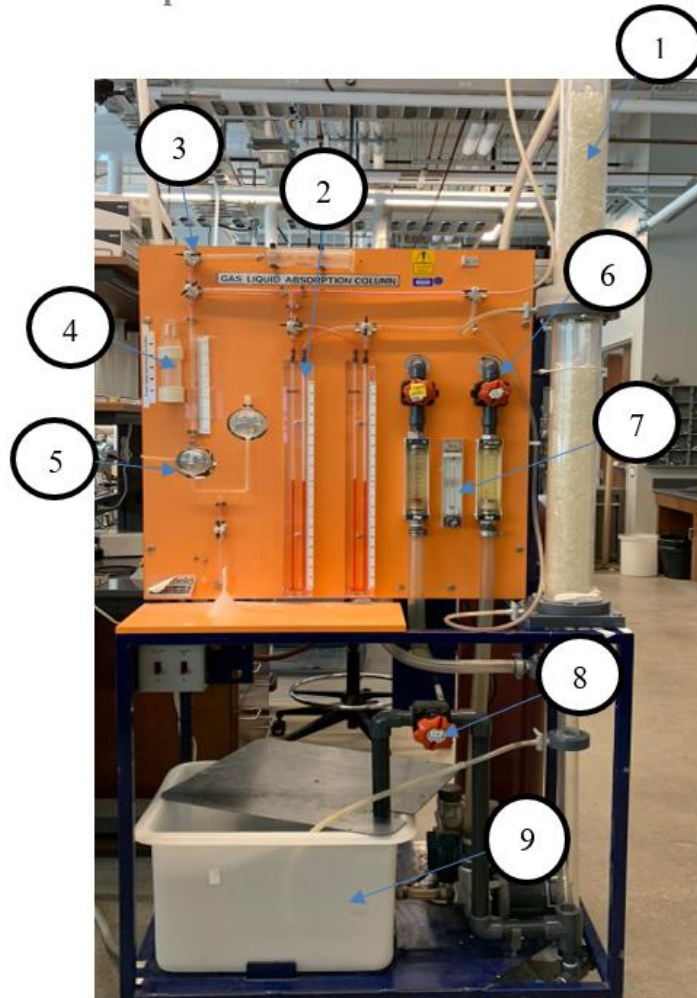
Where:

C_{CO_2} = Concentration of CO₂ in water

V_{base} = Volume of base V_{water} = Volume of water

C_{base} = Concentration of base

Apparatus



Apparatus

Number	Equipment	Description
1	Gas Absorption Column	Unit that is filled with the resins and has the flows applied to it
2	Manometer	Used to measure gas pressure
3	Manometer Control Valve	Controls the inlet of the gas pressure
4	Hempl piston	NaOH and CO ₂ piston used to measure mixture volume and concentration of CO ₂
5	Hempl container bulbs	Contains mixture of NaOH and CO ₂ to measure difference in mL
6	Water rotameter	Measures the flow rate of liquid or gas in closed tube
7	CO ₂ rotameter	Controls and measures the flow rate of CO ₂
8	Air rotameter	Controls and measures air flow rate in system
9	Water Reservoir Tank	Holds water in order to pump later in the apparatus
10	CO ₂ supply (heat block, regulator, and supply)	Regulates pressurized CO ₂ flow in system and has a heater block to prevent from freezing pipes
11	CO ₂ cylinder	Contains pressurized CO ₂
12	Water Pump	Sends water up to column

Material & Supplies

Materials/Supplies	Brief Description
Deionized Water	Water used to fill up the reservoir tank which fills up the column
Compressed Air	Used to flow through the column
Phenolphthalein Indicator Solution	Used for the titration indicator
0.05N Sodium Hydroxide Solution	Used for the procedures that follow the steps for Apparatus B
Carbon Dioxide Gas	Used to flow through the column
1.0N Sodium Hydroxide Solution	The base used for the titration standards
50mL burette and burette clamp	Used to add solution of NaOH to add into the CO ₂ samples
50mL graduated cylinder	Used to collect samples from the apparatus
Stopwatch	Used to maintain constant time intervals between each sample being collected
50mL pipet	Used to add the drops of the phenolphthalein indicator solution
125mL Erlenmeyer flask	Used for titration of the sample
Thermometer	To measure the temperature of the deionized water
Parafilm	To cover the samples of CO ₂ from evaporating
Stir Bars	To stir the components in the beaker
Stir Plate	To enable the stirring from the stir bar which is inside of the beaker
Bucket	To collect the water that is drained to maintain constant levels

Procedure

- Pre-Lab:

Section 1: Void fraction and determining Flooding for air/water flowrates

Void Fraction

- Two ways:
 - Density of solid packing
 - Amount of water needed to immerse a volume of packing
 - Calculate the void volume

Flooding for air/water flowrates

- Fill reservoir $\frac{3}{4}$ full of deionized water and record temp
- Set air flow rate to 20 L/min for 5 minutes and record pressure drop
- Next, set water flow rate to 1.5 L/min and increase air flow rate until flooding occurs
- Repeat this with different flow rates of 3, 6, and 8 L/min and increasing air flowmeter
- Record data

Procedure (cont.)

- Lab:

Section 2a: Equilibrium of water and CO₂

- Fill reservoir $\frac{3}{4}$ full of deionized water, record temp, and place lid on tank
- Collect sample to determine initial CO₂ concentration
- Start pump, adjust water flow to 6 L/min and air flowrate to 20 L/min
- Turn on CO₂ flowrate anywhere between 2-5 L/min and after 10 minutes, take 50 mL samples at 1 minute intervals (up to you) until steady state is reached (CO₂ concentration is constant in titration)

Section 2b: Desorption CO₂

- Turn off CO₂ flow and decrease water flowrate to 1.5 L/min, take samples in 1 minute intervals
- Titrate until steady state has been reached

Section 3: Absorption of CO₂

- Purge tank, then fill again $\frac{3}{4}$ full of deionized water, record temp, place lid on tank, and keep deionized water flowing in tank by carefully draining tank so it can be at steady state at 1.5 L/min. Collect sample to determine initial CO₂ concentration, once ready, allow CO₂ to flow
- Set air flowrate to 20 L/min and CO₂ flowrate between 2-5 L/min and immediately collect 50 mL samples at 30 second intervals for 2 minutes, 1 minute intervals for 5 minutes, and 5 minute intervals until steady state is reached in titration.

Experimental Challenges

- Make sure the reservoir tank has the drain closed while filling it up with water
- Make sure to cover the collected samples with parafilm as quickly as possible or else CO₂ will be lost in the sample
- Make sure to not overshoot titrations
- Make sure to turn back the CO₂ after part 2B
- Make sure to label flasks ahead of time to keep everything organizing while performing the procedure

Safety

- Make sure not to touch the heater
- Always wear gloves while dealing with the NaOH solutions and the indicator
- Make sure to place the water hose back into the sink after use
- Make sure to clean up any water that has fallen on the floor

References

- [1] Wankat, Phillip C. “Separation Process Engineering : Includes Mass Transfer Analysis.” *International Journal of Environmental Studies*, vol. 66, no. 6, 2012.
- [2] Riazi, M. R. (1996). Modeling of gas absorption into turbulent films with chemical reaction. *Gas Separation and Purification*, 10(1), 41–46.
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